

Solutions to Practice Exam Material, Part 1

① When $X_i \sim \text{Exponential}(\theta)$, we know that $E(X_i) = 1/\theta$. The first sample moment is $\bar{X}$. So, if we set

$$1/\theta = \bar{X}$$

we find $\hat{\theta}_{\text{mom}} = 1/\bar{X}$.

② C is correct.

③ A is correct. A more flexible model will necessarily have larger likelihood at its maximum.

④ C is correct.

⑤ $f_X(x; \theta) = \frac{1}{2\theta} \exp\left(-\frac{|x|}{\theta}\right), -\infty < x < \infty$

(This is the double exponential distribution.)

The joint distribution of $X_1, X_2, \dots, X_n$ is

$$\prod_{i=1}^n f_{X_i}(x_i; \theta) = \left(\frac{1}{2\theta}\right)^n \exp\left(-\frac{\sum |x_i|}{\theta}\right)$$

Hence the log likelihood is

$$l(\theta) = -n \log(2\theta) - \frac{\sum |x_i|}{\theta}.$$

$$\frac{\partial l(\theta)}{\partial \theta} = -\frac{n}{\theta} + \frac{\sum |x_i|}{\theta^2}$$

Set this equal to 0 to find

$$\hat{\theta}_{MLE} = \frac{\sum |x_i|}{n}.$$

(We note that $\frac{\partial^2 l(\theta)}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2\sum |x_i|}{\theta^3}$.
At $\hat{\theta} = \frac{\sum |x_i|}{n}$, this equals $\frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} < 0$.)

(6) a) We know that

$$\frac{\partial^2}{\partial p^2} f_X(x; p) = -\frac{X}{(1-p)^2} - \frac{1}{p^2}$$

$$\text{Hence, } I(p) = E\left(\frac{X}{(1-p)^2} + \frac{1}{p^2}\right)$$

$$= \frac{1}{p(1-p)} + \frac{1}{p^2}$$

$$\text{since } E(X) = \frac{1-p}{p}$$

$$= \frac{p + (1-p)}{p^2(1-p)}$$

$$= \frac{1}{p^2(1-p)}$$

$$\text{and } nI(p) = \frac{n}{p^2(1-p)}$$

Thus, $\hat{p}$ is approx. normal with
variance $\frac{p^2(1-p)}{n}$

$$\text{or SE } \sqrt{\frac{p^2(1-p)}{n}}$$

(b) We want to approx. the distribution of the MLE for $E(X_i) = \frac{1-p}{p}$

$$\text{Let } g(p) = \frac{1-p}{p}.$$

The MLE for $g(p)$ is $g(\hat{p}) = \frac{1-\hat{p}}{\hat{p}}$

by the invariance property.

We will use the Δ -method to approx. the dist. of $g(\hat{p})$.

We are given that

$$\frac{\partial g(p)}{\partial p} = -\frac{1}{p^2}$$

Hence, $g(\hat{p})$ is approx. normal

with approximate variance

$$\begin{aligned} & \left(g'(\hat{p}) \right)^2 \left(\frac{1}{nI(\hat{p})} \right) \\ &= \left(\frac{1}{\hat{p}^4} \right) \left(\frac{\hat{p}^2(1-\hat{p})}{n} \right) \\ &= \frac{(1-\hat{p})}{n\hat{p}^2} \end{aligned}$$

Thus, a $100(1-\alpha)\%$ CI for $E(X_i)$ is

$$\left(\frac{1-\hat{p}}{\hat{p}} \right) \pm z_{\alpha/2} \sqrt{\frac{(1-\hat{p})}{n\hat{p}^2}}$$

⑦ (a) Each X_i has pmf $f_{X_i}(x_i) = p^{x_i}(1-p)^{1-x_i}$

The likelihood is hence

$$L(p) = p^{\sum x_i} (1-p)^{n-\sum x_i}$$

The log-likelihood is

$$l(p) = (\sum x_i) \log p + (n - \sum x_i) \log(1-p)$$

$$\frac{\partial l(p)}{\partial p} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p}$$

Set this equal to 0 and find

$$\hat{p}_{MLE} = \frac{\sum x_i}{n} = \text{the sample proportion}$$

We note that $\frac{\partial^2 l(p)}{\partial p^2} = -\frac{\sum x_i}{p^2} - \frac{n - \sum x_i}{(1-p)^2}$

which equals

$$-\frac{n}{\hat{p}} - \frac{n}{(1-\hat{p})} < 0$$

when evaluated at $\hat{p}$.

The expected value of $\hat{p}$ is

$$E(\hat{p}) = E\left(\frac{\sum x_i}{n}\right) = \frac{\sum E(x_i)}{n} = \frac{np}{n} = p$$

(b) The exact variance of $\hat{p}$ is

$$\begin{aligned} V(\hat{p}) &= V\left(\frac{\sum X_i}{n}\right) = \frac{\sum V(X_i)}{n^2} \quad \text{since } X_i \text{ independent} \\ &= \frac{n p(1-p)}{n^2} = \frac{p(1-p)}{n} \end{aligned}$$

If we use Fisher Information,
we note that

$$\begin{aligned} nI(p) &= E\left(-\frac{\partial^2 \ell(p)}{\partial p^2}\right) \\ &= E\left(\frac{\sum X_i}{p^2} + \frac{n - \sum X_i}{(1-p)^2}\right) \\ &= \frac{n}{p} + \frac{n}{(1-p)} \\ &= \frac{n}{p(1-p)} \end{aligned}$$

So, this approach leads to a
variance of $\frac{p(1-p)}{n}$ as well

(c) Here we want to estimate $g(p) = \log\left(\frac{p}{1-p}\right)$

(i) The MLE for $g(p)$ is

$$g(\hat{p}) = \log\left(\frac{\hat{p}}{1-\hat{p}}\right)$$

by the invariance property.

(ii) We will use the Δ -method.

Note that

$$\frac{\partial g(p)}{\partial p} = \left(\frac{1-p}{p}\right) \frac{\partial}{\partial p} \left(\frac{p}{1-p}\right)$$

$$= \left(\frac{1-p}{p}\right) \left(\frac{(1-p) + p}{(1-p)^2}\right)$$

$$= \frac{1}{p(1-p)}$$

So, the variance of $g(\hat{p})$ is

Approximately

$$\left(\frac{1}{p(1-p)} \right)^2 \left(\frac{p(1-p)}{n} \right) = \frac{1}{np(1-p)}$$

which we would approximate using

$$\frac{1}{n\hat{p}(1-\hat{p})}$$

(iii) In this case,

$$\hat{p} = \frac{5}{10} = 0.5$$

So the approx. variance is

$$\frac{1}{(10)(0.5)(0.5)} = 0.4$$

$$\text{Also, } \log \left(\frac{\hat{p}}{1-\hat{p}} \right) = 0$$

So, the 95% CI is

$$0 \pm (1.96) \sqrt{0.4}$$

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(a) Since $X_1, X_2, \dots, X_n$ are iid $N(0, \sigma^2)$, we know that $E(X_i) = 0$.

In order to use the method of moments, the population moment(s) must be functions of the unknown parameters. So, we consider the second moment:

$$E(X_i^2) = V(X_i) + E(X_i)^2 = \sigma^2$$

The second sample moment is

$$\sum X_i^2 / n$$

Hence, the MOM estimator is $\hat{\sigma}_1^2 = \sum X_i^2 / n$

(b)

$$E(\hat{\sigma}_1^2) = E\left(\frac{\sum X_i^2}{n}\right) = \frac{\sum E(X_i^2)}{n} = \frac{n\sigma^2}{n} = \sigma^2$$

Yes, $\hat{\sigma}_1^2$ is an unbiased estimator for σ^2 .

$$(c) \quad \hat{\sigma}_2^2 = \frac{1}{n} \left(\sum X_i \right)^2 = n \bar{X}^2$$

$$E(\hat{\sigma}_2^2) = E(n \bar{X}^2) = n E(\bar{X}^2)$$

$$= n \left(V(\bar{X}) + E(\bar{X})^2 \right)$$

$$= n \left(\frac{\sigma^2}{n} + 0 \right)$$

$$= \sigma^2$$

Yes, $\hat{\sigma}_2^2$ is an unbiased estimator of σ^2 .

9 $f_X(x; \theta) = \theta e^x e^{-\theta e^x}, -\infty < x < \infty$

The likelihood function is

$$L(\theta) = \theta^n e^{\sum x_i} e^{-\theta \sum e^{x_i}}$$

and the log likelihood is

$$l(\theta) = n \log \theta + \sum x_i - \theta \sum e^{x_i}.$$

$$\frac{\partial l(\theta)}{\partial \theta} = \frac{n}{\theta} - \sum e^{x_i}$$

Solving for θ , we find

$$\hat{\theta}_{MLE} = \frac{n}{\sum e^{x_i}}$$

$$\left(\text{we note that } \frac{\partial^2 l(\theta)}{\partial \theta^2} = -\frac{n}{\theta^2} < 0. \right)$$

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Here, $f_X(x; \theta) = \frac{2x}{\theta} e^{-x^2/\theta}$

So the likelihood function is

$$L(\theta) = \frac{2 \prod_{i=1}^n x_i}{\theta^n} e^{-\sum x_i^2 / \theta}$$

and the log likelihood is

$$l(\theta) = \log(2 \prod_{i=1}^n x_i) - n \log \theta - \sum x_i^2 / \theta$$

$$\frac{\partial l(\theta)}{\partial \theta} = -\frac{n}{\theta} + \frac{\sum x_i^2}{\theta^2}$$

Setting this equal to 0, we find

$$\hat{\theta}_{MLE} = \frac{\sum x_i^2}{n}$$

(We note that $\frac{\partial^2 l(\theta)}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2 \sum x_i^2}{\theta^3}$

which equals $\frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} < 0$
at $\hat{\theta}$.)

(b) First, find that, for $t > 0$,

$$\begin{aligned} P(X > t) &= P(X^2 > t^2) \\ &= e^{-t^2/\theta} \end{aligned}$$

Using the given fact that
 $X^2 \sim \text{exponential}(1/\theta)$.

(Clearly, for $t \leq 0$, $P(X > t) = 1$.)

(c) We know that

$$\frac{\partial^2 l(\theta)}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2 \sum X_i^2}{\theta^3}$$

$$\text{So, } nI(\theta) = E \left[\frac{2 \sum X_i^2}{\theta^3} - \frac{n}{\theta^2} \right]$$

$$= \frac{2n\theta}{\theta^3} - \frac{n}{\theta^2}$$

Since
 $E(X_i^2) = \theta$

$$= n/\theta^2$$

Hence, $\hat{\theta}_{MLE}$ is approx. normal
with variance θ^2/n .

(d) A $100(1-\alpha)\%$ CI is

$$\hat{\theta} \pm z_{\alpha/2} \left(\frac{\hat{\theta}}{\sqrt{n}} \right)$$

(e) $g(\theta) = \sqrt{\theta}$.

The MLE for $g(\theta)$ is $\sqrt{\frac{\sum X_i^2}{n}}$

by the invariance property.

We know that $\hat{\theta}_{MLE}$ is an unbiased estimator of θ , because $E(X_i^2) = \theta$.

Unfortunately,

$$E(\sqrt{\hat{\theta}}) \neq \sqrt{E(\hat{\theta})} = \sqrt{\theta}$$

Hence, $\sqrt{\hat{\theta}}$ is not an unbiased estimator for $\sqrt{\theta}$.

(f) We will use the Δ -method.

We note that

$$\frac{\partial g(\theta)}{\partial \theta} = \frac{1}{2\sqrt{\theta}}$$

So the approx. variance is

$$\left(\frac{1}{2\sqrt{\theta}}\right)^2 \left(\frac{\theta^2}{n}\right) = \frac{\theta}{4n}$$

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$$f(x; \theta) = \theta e^x e^{-\theta e^x}, \quad -\infty < x < \infty$$

(a) The likelihood function is

$$L(\theta) = \theta^n e^{\sum x_i} e^{-\theta \sum e^{x_i}}$$

and the log likelihood is

$$l(\theta) = n \log \theta + \sum x_i - \theta \sum e^{x_i}.$$

$$\text{Then, } \frac{\partial l(\theta)}{\partial \theta} = \frac{n}{\theta} - \sum e^{x_i}.$$

Setting this equal to 0, we find

$$\hat{\theta}_{MLE} = n / \sum e^{x_i}.$$

(We note that

$$\frac{\partial^2 l(\theta)}{\partial \theta^2} = -\frac{n}{\theta^2} < 0$$

for any θ .)

(b) We know that

$$nI(\theta) = E\left(-\frac{\partial^2 \ell(\theta)}{\partial \theta^2}\right) = \frac{n}{\theta^3}$$

So, the approx. SE is $\frac{\hat{\theta}}{\sqrt{n}}$.

(12) (a) Yes, it is unbiased since

$$E(\bar{X}) = \lambda$$

$$(b) V(\bar{X}) = \frac{\sum V(X_i)}{n^2} = \frac{n\lambda}{n^2} = \frac{\lambda}{n}$$

$$\text{so, } \text{MSE}(\bar{X}) = \lambda/n$$

$$\text{since bias}^2 = 0.$$

$$(c) 7.2 \pm (1.96) \sqrt{\frac{7.2}{50}}$$

(13) The 90% CI is

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}} \right) = \left(\frac{(99)(79.3)}{\chi^2_{.05, 99}}, \frac{(99)(79.3)}{\chi^2_{.95, 99}} \right)$$

$\alpha/2$ $1-\alpha/2$.05 .95

(14)

The MLE for μ is $\bar{X}$

The MLE for σ^2 is $\frac{n-1}{n} s^2$

Hence, by the invariance property,
the MLE for μ/σ is

$$\frac{\bar{X}}{\sqrt{\left(\frac{n-1}{n}\right) s^2}}$$

(15)

We can use the Δ -method.

Let $\theta = \sigma^2$. We want approx.
variance of $g(\theta) = \sqrt{\theta} = \sigma$.

$$\frac{\partial g(\theta)}{\partial \theta} = \frac{1}{2\sqrt{\theta}}$$

So, the approx. variance of $g(\hat{\theta}) = \hat{\sigma}$ is
 $\left(\frac{1}{2\sqrt{\theta}}\right)^2 (2\theta^2/n) = \theta/2n = \sigma^2/2n$.